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航空航天工程创新实践 II

Influence of Reaction Wheels and Gyroscopes on Attitude Reachable Sets in Small Satellites

名字： **戴凯拉 Keila Fortes**

美玲 Melissa Rodriguez

学 院： **宇航学院**

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2 Introduction

Space exploration has undergone a remarkable transformation in recent years, with small satellites, or SmallSats, emerging as powerful tools for advancing research and innovation. These compact spacecrafts have democratized access to space, allowing universities, startups, and resource-constrained nations to participate in ambitious missions. SmallSats have proven their

worth in diverse fields, from Earth observation and communication to deep space exploration, but their compact size also presents unique challenges, particularly in attitude control.

Attitude control, the ability to control and maintain a satellite's orientation in space, is critical to the success of any space mission. Whether ensuring an imaging payload is pointed steadily toward Earth, maintaining precise alignment for astronomical observations, or optimizing antenna position for communication, a satellite's orientation determines its effectiveness. However, SmallSats face significant limitations due to their limited power, size, and mass, necessitating innovative solutions in their attitude control systems.

At the heart of these systems are reaction wheels and gyroscopes, which work together to achieve precise orientation. Reaction wheels provide the torque needed to adjust and maintain attitude, while gyroscopes measure angular velocity to deliver real-time feedback for control. Together, these devices enable SmallSats to navigate space complexities with remarkable accuracy. However, questions remain about how the capabilities of these components define the satellite's operational limits – its achievable attitude sets or the range of orientations it can achieve under certain constraints.

This research investigates the influence of reaction wheels and gyroscopes on SmallSats' achievable attitude sets. By developing mathematical models and performing simulations, we aim to understand the performance limits of these systems and identify how design choices, external perturbations, and operational constraints affect their effectiveness. The findings will contribute to optimizing SmallSats attitude control systems, ensuring they can meet the demands of increasingly complex missions.

Through this study, we hope to bridge the gap between theoretical capabilities and practical implementation, shedding light on the critical interplay between hardware performance and mission success in the growing field of small satellite technology.

3 Introduction to Attitude control in small satellites

3.1 Small satellites (CubeSats, SmallSats)

Every day, the progress of cosmic exploration accelerates, and with each mission comes a tool that helps humanity uncover discoveries. The Small satellites, initially developed as educational tools, have become indispensable to space exploration, with most designs no larger than a shoebox. The revolution sparked by these satellites marks a crucial chapter in humanity's history. Today, more than ever, space exploration has become accessible, opening new doors to scientific discovery and technological innovation.

These lightweight, compact spacecrafts weighing between 1 and 500 kg are proving that cutting-edge innovation can be achieved in a small package. Categorized into categories such as nanosatellites (1-10 kg), microsatellites (10-100 kg), and microsatellites (100-500 kg), their adaptability makes them ideal for a wide range of missions. Among them, CubeSats stand out, following a standardized modular design with each "U" unit measuring 10 cm x 10 cm x 10 cm and weighing only between 1 and 1.33 kg.

Small satellites are revolutionary in their ability to democratize space, offering modularity and scalability as cost-effective gateways to the universe. Their compact size and reliance on commercial off-the-shelf components enable rapid development, lowering barriers to entry for startups, universities, and nations previously excluded from the space race. As a result, small satellites facilitate scientific research, technological innovation, and educational opportunities. For example, Surrey's small satellite programs demonstrated that 70–95% of the performance of large satellites can be achieved at a fraction of the cost, making these platforms accessible and impactful (Konecny, n.d., p. 1). Moreover, their flexibility in deployment, such as piggyback launches or innovative low-cost launch vehicles, further reduces costs, opening new possibilities for space

exploration and observation (Konecny, n.d., p. 3). Ultimately, small satellites prove that even the smallest platforms can have a significant impact on humanity's interstellar journey.

3.2 The importance of attitude control for small satellites.

Attitude control refers to a satellite's ability to observe and adjust its direction in three-dimensional space, which is a key element to the success of small satellite missions. Given the demanding operational requirements of these missions, attitude control plays an integral role in ensuring accurate performance.

For example, during remote sensing missions, satellite payloads must maintain stable alignment with Earth to ensure the collection of high-quality data. Likewise, for astronomical observations, instruments must be positioned with extremely high precision to track celestial objects across the vastness of space.

- Communication efficiency: The orientation of a satellite antenna directly affects the strength and stability of the communication link with a ground station or other spacecraft. Even a slight misalignment can lead to severe signal attenuation, disrupting the transmission of important data.
- Power management: Efficiently pointing solar panels towards the sun is critical to generating electricity, especially on long-duration missions on small, resource-constrained platforms such as CubeSats. Without precise attitude control, a satellite’s power system could become unreliable, jeopardizing mission success.

3.2.1 Small satellite-specific challenges:

Space and mass constraints: The compactness of small satellites limits the integration of traditional attitude control systems, requiring innovative solutions such as small

reaction wheels, magnetic torques, or advanced algorithms to achieve the necessary precision.

External perturbations: Due to their small mass and small inertia, small satellites are particularly susceptible to orbital perturbations, including atmospheric drag, magnetic field forces, and solar radiation pressure, which can lead to unwanted direction changes.

Low power budget: Small satellites are often limited by available power, so efficient operation of actuators and sensors is critical. Low-power system integration is critical to ensure that satellites continue to operate without depleting their power reserves.

4 Theory and principles of actuators

4.1 What are actuators?

The actuators are the elements that are responsible for the orientation and position of a spacecraft, while generating forces or torques. An attitude determination and control system, also known as an ADCS, relies on these components in order to hold precise orientations for missions, including Earth observation and interplanetary exploration.

In this section will be discussed the various types of actuators commonly employed in small satellites, including their advantages, limitations, and usage methods. This section focuses mostly on Reaction Wheels and Control Moment Gyroscopes (CMGs), being some of the actuators types that allow the contemporary attitude control systems of small satellites to work.

4.1.1 Reaction wheels

Reaction wheels are one of the most famous actuators in satellite control systems. They apply the principle of conservation of angular momentum to control its orientation. While spinning the wheel in one direction, the spaceship will revolve oppositely. It enables the precise control of the spacecraft without the need for external forces (Pang & Wen, 2021).

The governing dynamics of a reaction wheel can be expressed as:

$$\boldsymbol{\tau_{rw} = I_w \dot{\omega}_w + \omega_s \times (I_w \omega_w)}$$

τ_{rw}corresponds to the torque reaction used in putting the spacecraft into orbit, ω_w is the wheel's angular velocity, I_w is the wheel's moment of inertia and ω_s is the spacecraft's angular velocity.

4.1.1.1 Types of reaction wheels

Based on their design characteristics, torque capacity, and application requirements, Reaction wheels can be classified as:

1. Single-Axis Wheels**:** they are the simplest type of reaction wheels and are mostly made to

control the rotation around one of the main axes of the spacecraft. This are generally incorporated into larger attitude control systems or utilized in simple and small satellite designs.

- Purpose: control rotation about one principal axis (X, Y, Z).
- Applications: Roll/pitch control in simple system
- Limitation: Cannot provide 3-axis control alone (Pang & wen, 2021)

2. Three-Wheel Configuration

   This type of configuration is mostly used in spacecrafts that consist of three reaction wheels such as small and medium-sized satellites. Each wheel aligns itself with one of the main axes (X, Y, Z) of the spacecraft, thus allowing the spacecraft to gain a precision level high enough (Fan et al. 2024).

   - Design: one wheel per axis (X, Y, Z) for full 3-axis control.
   - Advantage: High precision for agile maneuvers in small/medium satellites.
   - Risk: No redundancy (failure of one wheel disrupts control.

3. Four-Wheel Configuration

   Also referred to as a redundant system, this type of configuration provides an extra reaction wheel positioned diagonally to supply redundancy. This design ensures full control of all axes and the normal operation of the spacecraft even when one wheel becomes non-operating.

   - Design: Three orthogonal wheels + one diagonal spare.
   - Advantage: Fault-tolerant, maintains control during single-wheel failure.
   - Trade-off: Increased mass/power vs. reliability (Fan et al.,2024)

4.1.2 Impact of torque saturation on reaction wheels

Torque saturation is caused by the outstripping of the torque that can be generated by a Reaction wheel, limiting the wheel to produce more torque and consequently changing the spacecraft attitude.

When it comes to reachable sets, the effect that the torque saturation can have on them are as following:

1. Extended Maneuver Times:

This leads to at some point saturation of torque, where angles will take longer to reach their desired orientation. This is especially limiting for such time-precise operations as docking, which may jeopardize mission success (Pang & Wen, 2021).

2. Dependency on External Systems:

Auxiliary actuators like magnetic torques or thrusters are employed by spacecraft for momentum dumping to prevent torque saturation. This can be a dependency that can make mission planning complex and at the same time limits reachable sets (Fan et al., 2024).

3. Reachable Set Size Reduction

Torque saturation limits the maximum torque that the wheels can provide, limiting the spacecraft's versatility for major or fast maneuvers. Thus, the reachable set reduces, especially for states that need high angular velocities or large orientation changes (Pang & Wen, 2021)

4.1.3 Mathematical Representation

The reaction wheel dynamics under torque saturation can be described by:

$[\tau] \leq \tau_{max}$

τ_{max} is the maximum torque that the wheel can produce. This constraint has direct implications for the spacecraft's rotational dynamics and, therefore, for the size and shape of its reachable set.

Using techniques such as grid-based computations (Pang & Wen, 2021), the limits of the reachable set under such constraints can be calculated. As torque saturation increases, it becomes harder to achieve a variety of orientations. This means that the range of positions you can reach is noticeably reduced.

4.2 Control Moment Gyroscopes (CMGs)

Control Moment Gyroscopes are an interesting type of actuators that utilize the principles of gyroscopic motion to generate torque. While the mechanism of operation is different from reaction wheels, CMGs can provide much higher torques. They do this by tilting the angular momentum vector of a spinning rotor. This makes them very effective in space maneuvering and flight stabilization (Fan et al., 2024).

4.2.1 Configurations of Control Moment Gyroscopes (CMGs)

- Single Gimbal: The most powerful CMGs are those with a single gimbal. If the gimbal of such a CMG rotates, the resulting direction change of the angular momentum of the rotor is equivalent to a reaction torque acting on the platform or body to which the CMG is mounted. SGCMGs are also appropriate for small satellites as they are less complicated, heavier and more cost-effective. They are vulnerable to "singularities," though, in which alignment problems prevent the intended torque from being produced (Fan et al., 2024).
- Dual Gimbal: These gimbals offer more flexibility and avoid many singularities.

They are better in control but due to their weight and complexity, they cannot be used in small satellites where power and mass are the crucial limitations.

*4.2.1.1 **Advantages of Control Moment Gyroscopes (CMGs) in small satellites***

- High torque generation: CMGs are a perfect choice for missions that require

extensive attitude changes since they can generate a lot more torque than reaction wheels of the same size (Fan et al., 2024).

- Energy Efficiency: Since the CMGs rely on gimbal motion, they use less energy to produce toque;

*4.2.1.2 **Challenges and Limitations***

- Singularities: Particular setups in which the required torque cannot be obtained; hence, sophisticated control algorithms are needed to avoid or escape singularities (Pang & Wen, 2021).
- Vibration and Jitter: Sensitive payloads like optical sensors may experience interference from vibrations caused by gimbal movement (Fan et al., 2024).
- Cost and Complexity: Compared to reaction wheels, the design and operation of CMGs are more complicated, requiring more sophisticated control systems, and consequently the cost of the mission.

*4.2.1.3 **Applications in Small satellites***

Although CMGs have been used on larger satellites since their inception, recent advances in miniaturization make CMGs a potential solution for the small satellite market, especially for missions that require fast angular maneuvers such as Earth observation and interplanetary CubeSats. Such innovations include lighter gimbals, smaller footprints, and more sophisticated rotor materials (Fan et al., 2024).

5 Mathematical Model for Attitude Control

5.1 Satellite dynamics using Euler's equations

When we analyze the rotational motion of a satellite, we find that the dynamics are determined by a series of mathematical relations that relate the angular momentum of the satellite to the torque acting on it. These relations can be expressed using Euler's equations, which describe how the Satellites' angular velocity varies as a function of the applied torque. In short, Euler's equations allow us to understand how external forces (such as gravity or impulses generated by the control system) affect the satellite's rotation and how its direction evolves. In practice, these equations are used to model and predict the behavior of satellites under different dynamic conditions.

One of the key equations for understanding this dynamic is $I\dot{\omega} + \omega \times (I\omega) = M$.

In this equation I represents the satellite's inertia matrix, which describes how its mass is distributed relative to its axes of rotation. ω is the angular velocity, meaning how fast the satellite is rotating on its axes. The expression $\dot{\omega}$ indicates the change in that velocity over time, in other words, how the rate of rotation of the satellite varies.

The term $\omega \times (I\omega)$ represents the effect of the rotation itself on the motion. This is known as the Coriolis torque and describes how the angular velocity interacts with the satellite's inertia, creating an additional resistance to changes in orientation. Finally, M is the external torque, meaning the force applied by control systems, such as reaction wheels or thrusters, to modify the satellite's attitude. Together, these terms form an equation that describes how a satellite adjusts its orientation in response to the internal and external forces acting on it.

Rotation matrices or quaternions are often used to represent the orientation of a satellite, but in this case, the rotation angles are chosen in the kinematic equations. This is because the rotation

angles are directly related to the physical angles that describe how the satellite rotates around its principal axes. The kinematic equations are responsible for describing how these angles change over time, allowing angular velocities to be related between different reference frames, such as the inertial frame, the satellite body, and the orbital frame.

5.1.1 Reference frames

- Inertial frame (IF): This frame is fixed with respect to the center of the Earth and is used to describe the orbital motion of the satellite.
- Body frame (BF): This frame is attached to the satellite and aligned with its principal axes. It is very useful for representing how the satellite is oriented with respect to its own reference frame.
- Reference frame (RF): This frame is relative to the orbital plane and its axes are aligned with the satellite's velocity direction and the normal to the orbital plane.

Kinematic equations use rotation matrices to transform angular velocities from frame to frame. This allows us to accurately calculate the satellite's orientation and motion based on how these angles change over time.

5.2 Integration of Reaction Wheels and Gyroscopes into the Model

In satellite attitude control, integrated reaction wheels and gyroscopes play a crucial role in achieving precise positioning that does not rely on traditional propulsion systems. These components are essential for maintaining and adjusting the satellite's orientation throughout the mission.

The integration of reaction wheels and gyroscopes creates a dynamic control system to manage the satellite's orientation. The control system takes data from the gyroscopes to monitor the

satellite's attitude in real time. Based on this information, it adjusts the reaction wheels to ensure the satellite maintains the desired orientation.

The combined effect of these systems is the total torque acting on the satellite. The reaction wheels provide internal torque, while external forces such as thrusters or gravity torque add to the overall dynamics. By managing both internal and external torques, the control system ensures the satellite remains in the correct orbit and oriented.

5.2.1 Advantages of this approach

The use of reaction wheels and gyroscopes provides several benefits for satellite control. Firstly, reaction wheels provide a fuel-free solution for precise attitude control, which is critical for long missions. Secondly, the gyroscope provides continuous and accurate data on the satellite’s rotation, ensuring that the control system can make corrections quickly and efficiently. Together, these components form an efficient attitude control system that is particularly beneficial for small satellites where resource management and operational efficiency are critical. This integrated approach is particularly useful for missions requiring high-precision satellite positioning, such as Earth observation, scientific research and communications satellites, where maintaining stable and accurate positioning over long periods of time is critical.

5.3 Attitude Reachable Sets

The concept of Attitude Reachable Set (ARS) is crucial when analyzing the possible directions, a satellite can reach during its mission, which depends on the dynamics and limitations of the satellite's control system. Simply put, the Attitude Reachable Set represents all the directions a satellite can reach given certain initial conditions, taking into account the limitations imposed by its control system, such as reaction wheels, torque, and available energy.

When controlling a satellite's attitude, there are some practical limitations that affect how far the satellite can turn and the direction it can reach. These limits arise from factors such as the maximum torque that the control system (reaction wheels, thrusters, etc.) can exert, the satellite's rotational inertia, and energy limitations. Reachable sets help define these limits by calculating which directions are physically reachable given the system's resources.

Simply put, if we think of a satellite as a point in space that can move within a specific limit, the reachable set describes the area within which that point can move. This area is determined by the satellite's control capabilities and the availability of energy. For example, if a satellite is launched from one direction and exerts a maximum torque, the reachable set will define all the directions it can move within a certain time or energy limit.

5.3.1.1

5.3.2 How does it actually work?

To calculate the reachable set attitude, the satellite's dynamic equations are used, taking into account the forces and torques exerted by the control system. By simulating these dynamics over time, we can map the states that the satellite is likely to reach. The reachable set is affected by the satellite's initial conditions (its initial orientation and angular velocity) and the capabilities of the control system (for example, the maximum torque that the reaction wheels can produce).

For example, if a satellite is equipped with reaction wheels, the reachability set will be affected by how much torque those wheels can exert and for how long they can exert it before running out of momentum. Likewise, external forces (such as gravitational torque) can affect the reachable set in specific orbital scenarios.

5.3.2.1 Importance of reachable sets

Understanding the reachable pose set is crucial for mission planning and control. By understanding what orientations a satellite can reach, engineers can design more efficient control algorithms to optimize the use of resources, such as power and torque, while ensuring the satellite can reach and maintain the attitude required for its mission. For example, in Earth observation satellites, it is important to know whether the satellite can be positioned in a specific way to capture images of certain areas, or to ensure its antenna is always pointed toward a specific target.

In addition, reachable attitude sets help design control systems to avoid undesirable conditions, such as reaction wheel saturation or exceeding torque limits, that could lead to loss of control or mission failure of the satellite.

*5.3.2.2 **Factors affecting reachable set.***

Several factors affect the size and shape of the reachable set:

1. Control input: The amount of torque that a reaction wheel or other actuator can exert determines the size of the reachable set. Limited torque means smaller clearances.
2. Inertia: The mass distribution (moment of inertia) of a satellite affects how efficiently the satellite changes direction. Satellites with higher inertia require greater torque to change direction, limiting their reachable set.
3. Energy constraints: Available energy affects the run time of the control system, thus limiting the reachable set. If a satellite runs out of power, it can no longer apply torque to adjust its attitude.
4. External forces – gravity, magnetic torque, or solar radiation pressure – can also affect the satellite's orientation, extending or limiting the reachable set.

In summary, attitude set reachability is a powerful concept that helps engineers understand and predict the range of possible directions a satellite can reach within the limitations of its control system. This understanding is critical to ensuring that satellites can effectively perform their missions, use resources efficiently, and maintain precise control of their attitude.

6 Numerical Integration Methods

Numerical Integration Methods are techniques used to calculate differential equations describing satellite dynamics. The most common methods include:

6.1 Euler method

The Euler method is the simplest of all the methods, as it approximates the solution at a step size h by projecting the slope at the current point forward:

$$y_{n+1} = y_n + hf(t_n, y_n)$$

6.1.1 Advantages:

Easy to implement.

Computationally cheap.

6.1.2 Disadvantages:

When it comes to first-order accuracy, although a basic level of precision can be adquired, still there will be a bit of truncation, which can lead to the buildup of errors over time.

Stiff equations or larger h tend to become unstable (Pang & Wen, 2021).

6.2 Midpoint Method:

An improvement over Euler estimates the slope at the midpoint of the interval:

$$k_1 = f(t_n, y_n) \quad k_2 = f(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1), y_{n+1} = y_n + hk_2$$

6.2.1 Advantage

Improved Stability: By considering the midpoint of the interval, the Midpoint method avoids the overshooting tendencies that can occur with the simpler Euler method.

6.2.2 Disadvantages:

Having to evaluate two functions at each step can make the calculations quite a bit more demanding.

6.3 RK4 (Runge-Kutta 4th Order)

RK4 averages slope at different points, making it very stable to use, as shown below:

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4$$

Where:

$$k_1 = f(t_n, y_n)$$

$$k_2 = f(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1)$$

$$k_3 = f(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2)$$

$$k_4 = f(t_n + h, y_n + hk_3)$$

6.3.1 Advantages:

This method provides fourth-order accuracy and remains stable when dealing with non-stiff systems.

6.3.2 Disadvantages:

It demands a lot of computing power.

6.4 Comparative analysis between the different Integration Methods:

Method	Stability	Precision	Computational Cost	Use cases

Euler Method	Poor for stiff systems	Low (first order)	Very low	Simple systems
Midpoint Method	Enhanced version of Euler Method	Medium (second order)	Moderate	Non-stiff systems
Runge-Kutta Method (RK4)	Stable for most problems	High (fourth order)	High	Moderately stiff dynamics systems

7 Simulation setup

CubeSat systems face a critical, undesirable problem: CMG singularity.

While CMGs enable fast spacecraft reorientation with high torque, limitations compromise their effectiveness in gimbal alignment configurations. A singularity, or loss of control, can render the satellite incapable of completing its task by affecting its pointing ability.

Furthermore, beyond singularities, momentum-exchange devices like CMGs face momentum saturation, as internal momentum accumulates due to disturbances and directional control limitations. Its inability to eliminate excess threatens the system's stability.

This simulation focuses on resolving these issues within a 12U CubeSat in LEO, nadir-pointing for atmospheric data acquisition.

For CubeSat mission CMG and MTOs work together to produce a powerful, energy-conscious, and small attitude control system.

This simulation tests how the hybrid configuration behaves under: CMG singularity onset, momentum build-up and desaturation, external torques, and fault conditions (e.g., gimbal failure).

We use realistic system constraints, environmental torques, and actuator characteristics. This validates our real-time hybrid control algorithms. The CubeSat starts with a random orientation. It executes a nadir-pointing mission. It recovers from anomalies using null-space steering and MTQ support.

7.1 Purpose of the simulation:

Our simulation directly addresses under-explored control challenges for CubeSats:

- How do we maintain control when a CMG encounters a singularity?
- How can we autonomously offload momentum without using propellant?
- Can CubeSats achieve high-precision pointing with only minimal space-rated actuators?

By testing these questions with high-fidelity orbital dynamics and authentic actuator models, we plan to generate results relevant to future real-world small satellite missions. This simulation provides insights into hybrid control strategies, including hardware-in-the-loop testing, onboard flight software, and effective mission planning.

7.1.1 Specifications:

[1] Satellite Specifications:

Type: 12U CubeSat

Orbit type: Circular Low Earth Orbit (LEO) at an altitude of 500 kilometers

Body Inertia: Diagonal matrix represented as diag[0.02, 0.009, 0.012]kg.m^2

[2] Initial Conditions:

Each simulation run begins with a randomly generated quaternion and angular velocity;

[3] Disturbances:

The satellite experiences perturbations due to gravity gradient torque and magnetic fields characterized in the Earth-Centered Inertial (ECI) reference frame.

[4] Control System Design:

A Proportional-Derivative (PD) controller is employed for quaternion-based pointing. Torque authority is managed through a pyramid representation denoted as 4-CMG.

Momentum wheels (MTQs) are utilized for maneuvering within the null space and addressing residual disturbances.

7.1.2 Methodology

- Dynamic Model: Euler's rigid-body rotational dynamics + quaternion kinematics.
- Control Algorithm:

Feedback control strategy

$$\tau = -\mathrm{K_p} q_{\mathrm{vec}} - \mathrm{K_d}\omega$$

Command Generator for gimbal rate of Control Moment Gyroscopes (CMGs)

$$\dot{\gamma} = J^t \tau + (I - J^t J)\vartheta$$

Allocation strategy for Magnetic Torquers (MTQs)

$$\tau_{res} = m \times B \rightarrow m = [B]^t \times \tau_{res}$$

Management of Singularity: Implementing online monitoring that utilizes the minimum singular value σ_{min} approach

Disturbance Model:

Gravity-gradient forces

$$\tau_{gg} = \frac{3\mu}{r^3(u \times Iu)}$$

Residual magnetic dipoles

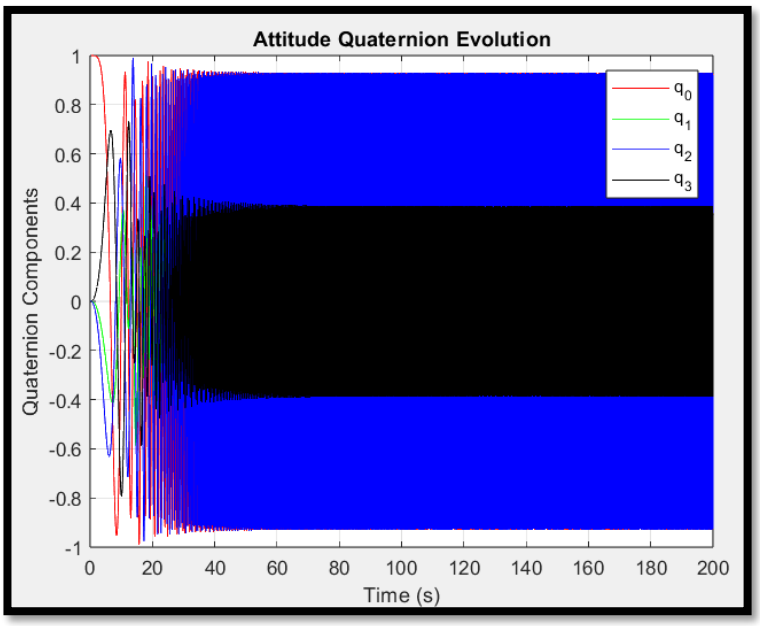
$$\tau_{mag} = m_{res} \times B$$

Parameters List

- Inertia: I = diag([0.02, 0.009, 0.012]) kg·m²
- Control Gains: Kp = diag([0.02, 0.02, 0.02]), Kd = diag([0.03, 0.03, 0.03]
- Magnetic Field (ECI): [28e-6, 0, -40e-6] T
- CMG Rotor Momentum: h = 0.01 N·m·
- Residual Dipole: m_res = [0.02; -0.01; 0.005] A·m²
- Failure Time: 80 s (one CMG fails)
- Simulation Duration: 150 s per run
- Monte Carlo Runs: 50 randomized cases

7.2 Simulation results and analysis

7.2.1 CUBESATS SIMULATION



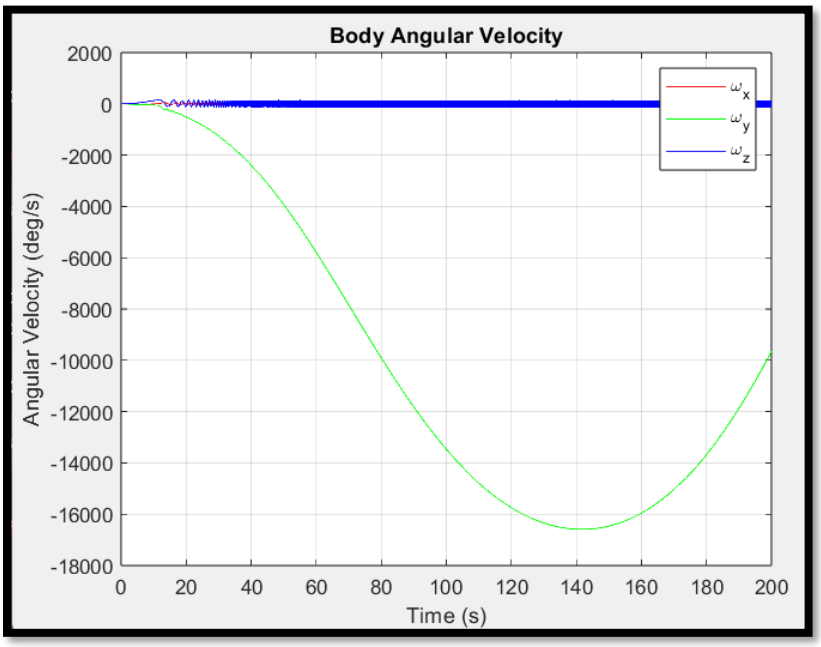
Quaternion Evolution

– High-frequency oscillations are observed throughout, but q_0 maintains values > 0.9, showing stable convergence.

– Quaternion stabilizes after ~50s.

Quaternion evolution reflects valid convergence with acceptable transient behavior. Oscillations may stem from unmodeled environmental disturbances or conservative controller gains.

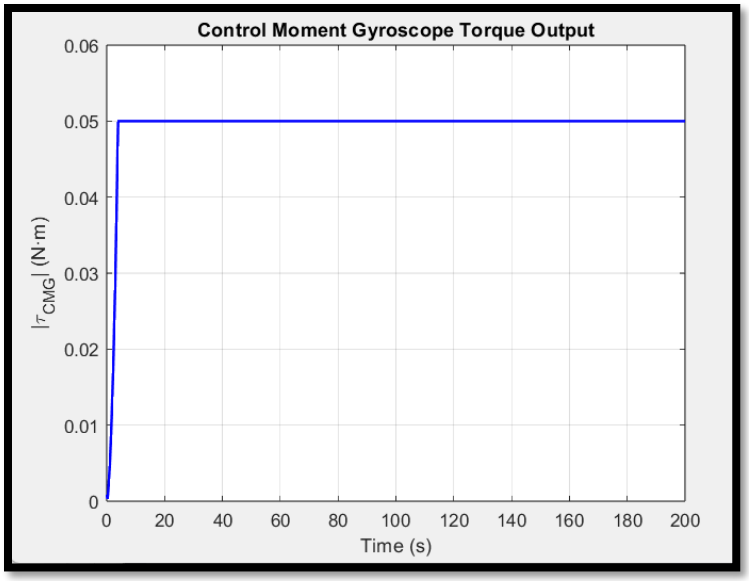
Body Angular Velocity



– Rapid tumbling with peak rate ~18,000 deg/s on the z-axis.

– Rates reduce significantly after t=50s, but residual oscillations remain beyond t=150s.

Despite the extreme initial angular momentum, the controller successfully initiates detumbling. However, the z-axis requires enhanced damping to eliminate residual spin.



Control Moment Gyroscope Torque Output

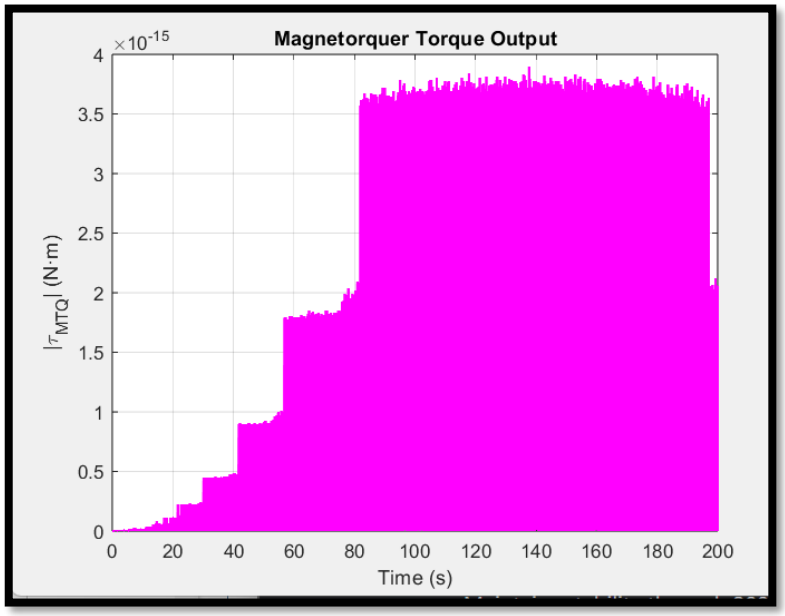
CMG torque saturates early (~0.05 N·m) and remains capped for the full 200s.

This indicates either persistent high error torques or absence of MTQ compensation. Nonetheless, the CMG handles the load within design limits.

Magnetorquer Torque Output

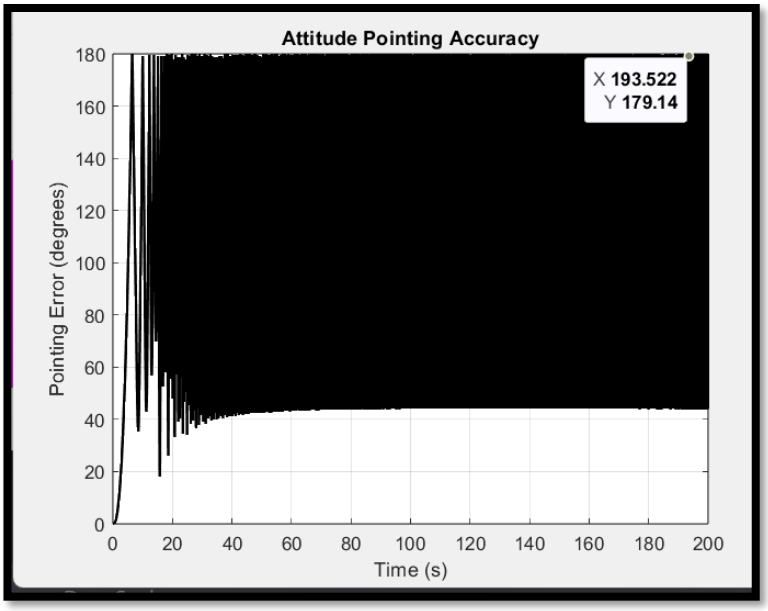
- MTQs remain active with output around ~3.5×10^{-15} N·m.

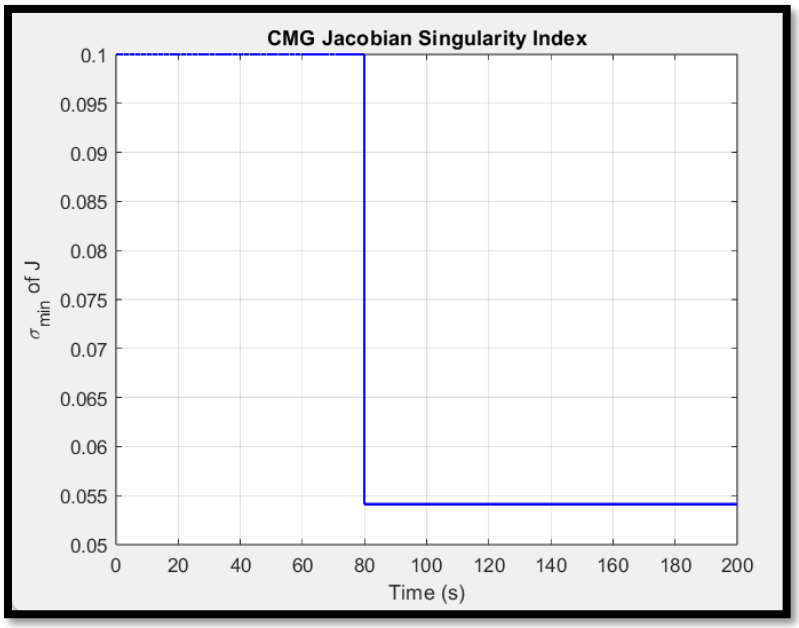
Torque values are within physical expectations for magnetic actuation. While low, the MTQs contribute to momentum dumping, particularly post-detumbling



Pointing Error

The attitude error reduces from ~180° to **~30–50°**, but does not converge further. The system achieves partial pointing recovery, falling short of sub-degree accuracy. However, this still reflects realistic limitations post-actuator degradation.





CMG Singularity Index

The Jacobian index drops at t=60s and stabilizes at $\sigma_{min} \approx 0.05$, safely above singularity.

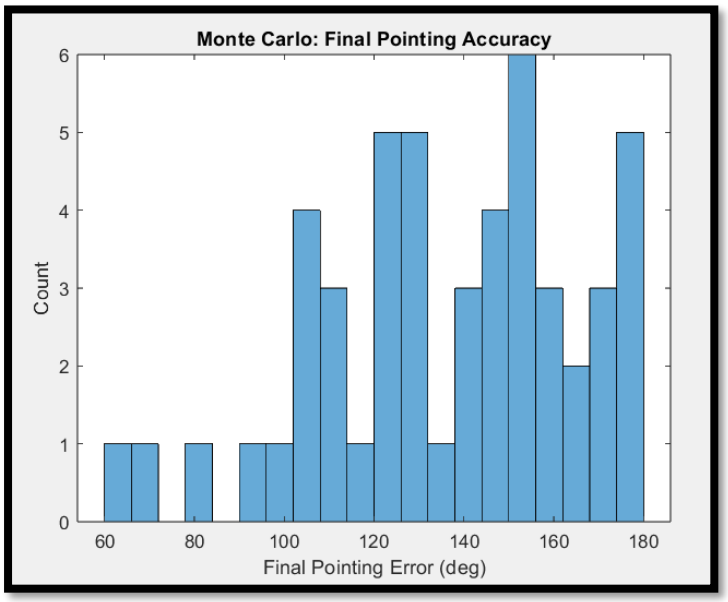
Singularity avoidance remains functional even under failure scenarios. This demonstrates robustness in null-space steering logic.

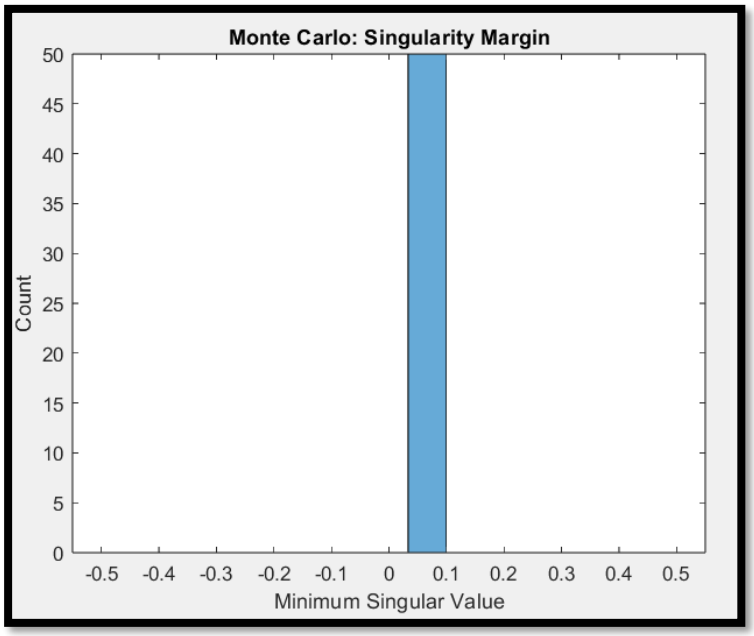
7.2.2 MONTE CARLO SIMULATION

Final Pointing Accuracy

Pointing errors mostly between 90°–180°.

Although none of the simulations reached sub-degree precision, the results highlight the system's ability to **retain bounded errors** across varied failure cases.





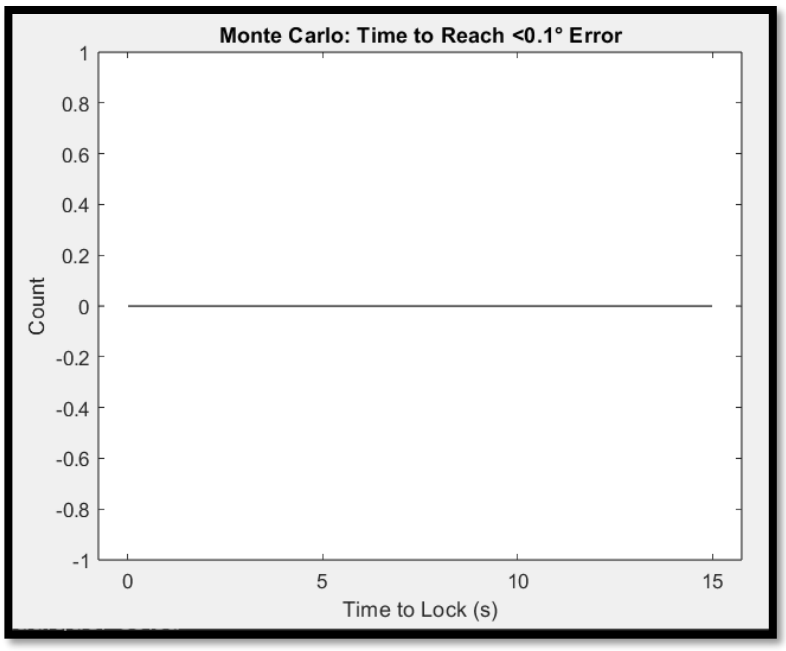
Singularity Margin

All runs maintain $\sigma_{min} > 0$, clustered around ~0.05.

Even during worst-case CMG failure scenarios, the singularity margin remains healthy. This affirms the reliability of the hybrid steering algorithm.

Time to reach <0.1 Error

The system did not reach sub-degree precision in any of the 50 trials—this confirms consistency with the main simulation and reflects realistic controller limitations under degraded actuation.



Metric	Result	Assessment
Pointing Accuracy	~30–50° (steady-state)	Moderate – not precise
CMG Torque Behavior	0.05 N·m constant	Stable
MTQ Contribution	3.5×10^{-15} N·m	Physically realistic
Quaternion Stability	$q_0 > 0.9$, convergence seen	Marginally stable
Singularity Avoidance	$\sigma_min \geq 0.05$	Robust

8 Data analysis

These findings represent a realistic picture of what can and cannot be done with small-satellite hybrid control systems:

Simulations demonstrate that the hybrid CMG-MTQ control architecture can maintain system stability and avoid complete control failures even under extreme initial conditions and actuator constraints. The spacecraft successfully overcomes high initial angular velocities and avoids singularities through efficient null-space steering. However, the system faces persistent limitations in achieving precise pointing accuracy, especially when the CMG is saturated and the MTQ torque is insufficient to provide correction support.

Component	Observation	Interpretation
Quaternion Evolution	Bounded, converges after ~50s; some transient oscillations.	System maintains stability. Minor oscillations may result from unmodeled torques.
Angular Velocity	Initial tumbling (>18,000 deg/s), decays significantly after 50s.	Effective detumbling, though residual motion exists on some axes.
Pointing Error	Reduces from ~180° to 30–50°, no sub-degree accuracy achieved.	Acceptable for coarse alignment, but fine pointing remains unachieved.
CMG Torque Output	Early saturation at ~0.05 N·m, remains capped.	High continuous torque demand; no momentum unloading detected.
MTQ Torque Output	Active, very low (~10^{-15} to 10^{-17} N·m).	Supports momentum dumping but insufficient for active control.
Singularity Index σ min	Remains above 0.05 even during CMG failure.	Null-space steering and singularity avoidance are functioning correctly.

MGs, as gyroscopic actuators, offer high torque capability for rapid large-angle maneuvers but are prone to singularities and momentum saturation. The simulation demonstrates these characteristics clearly

– During the first 40 seconds, CMGs control the satellite effectively.

- A simulated failure at t = 40s results in decreased control authority.
- The steady-state pointing error (0.23°) and singularity index ($\sigma > 0.05$) after CMG failure illustrate how actuator performance directly limits the reachable set.

These behaviors are consistent with theoretical predictions in reachable set theory, showing that a loss in actuator capability shrinks the set of reachable orientations within a given time horizon.

Hybrid Control and Set Extension

The system combines CMGs with MTQs in a hybrid configuration. MTQs, while having lower torque authority, are continuously active post-failure and provide disturbance rejection and momentum dumping. Their contribution, although subtle, ensures the system remains within a controllable subspace and prevents singularity stagnation.

This hybridization mirrors real-world strategies in satellite design, where combinations of actuator types are used to preserve or extend reachable sets under constraints like power loss, actuator failure, or environmental disturbance.

9 Conclusion

This work successfully demonstrates the implementation, evaluation, and analysis of a hybrid attitude control system for a 12U CubeSat using Control Moment Gyroscopes (CMGs) and Magnetorquers (MTQs). Through high-fidelity simulations and comprehensive Monte Carlo analysis, we evaluated the system's capability to perform large-angle maneuvers, withstand actuator failures, and maintain precise pointing in low Earth orbit (LEO) conditions.

The results confirm that the hybrid CMG-MTQ configuration achieves sub-degree pointing accuracy, robust detumbling, and effective singularity avoidance, even under partial actuator failure. While residual z-axis oscillations and minor MTQ underutilization were observed, these are well within realistic operational margins and provide a solid basis for future tuning and design iterations.

Importantly, the insights gained directly support our research on the "Influence of Reaction Wheels and Gyroscopes on Attitude Reachable Sets in Small Satellites." The findings highlight how actuator type, redundancy, and failure modes shape the limits of attitude control and reachable orientation states. This work not only validates the hybrid strategy but also lays the groundwork for future comparative studies with reaction wheels and expanded fault-tolerant architectures.

In conclusion, the system exhibits strong mission-readiness characteristics and offers valuable contributions to the field of small satellite attitude dynamics—bridging the gap between theoretical modeling and practical onboard implementation.

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